



COM 122, 119: Introduction to Programming and Object-oriented Programming

American University of Central Asia
Software Engineering Department

1 Course Description

This two-semester course provides a foundation in programming, emphasizing structured, procedural, and object-oriented paradigms in C++. Offered by AUCA's Software Engineering Department, it is a prerequisite for many departmental courses. It is also suitable for students from other departments who have no prior programming experience and want a general introduction to the field.

Students will learn essential concepts, including memory management, control flow, functions, procedural decomposition, objects, classes, encapsulation, inheritance, and polymorphism. The course emphasizes hands-on practice in developing C++ applications while building proficiency with fundamental development tools: code editors and integrated development environments (IDEs), compilers, build systems, package managers, debuggers, and version control systems. By the end of the course, students will be able to write efficient, well-structured programs and to apply good programming practices consistently.

2 Course Details

Course Materials

<https://github.com/auca/com.122-119>

Course Codes

COM 122

COM 119

Course IDs

5682

4357

Prerequisites

COM 122:

None

COM 119:

COM 122, Introduction to Programming

Prerequisites for

The list below depends on the student's year of admission.

COM 122:

COM 119, Object-oriented Programming

COM 123, Principles of Computing Systems

COM 119:

COM 213, Database

COM 223, Algorithms and Data Structures

COM 229, Data Structures

COM 421, Software Engineering I

COM 431, Senior Thesis I

MAT 407, Numerical Methods

Various elective courses

Credits

6

Professors, Time, Place**Dmitrii Toksaitov**

Lecture: Monday 14:10–15:25, CH 440

Lab: Wednesday 14:10–15:25, Lab 432

Adilet Abdykerimov

Lab: Wednesday 10:50–12:05, Lab G31

Lab: Wednesday 12:45–14:00, Lab G31

Lab: Wednesday 14:10–15:25, Lab G31

Lab: Wednesday 15:35–16:50, Lab G31

3 Contact Information

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TAs**Luna Maltseva**

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Office

AUCA, Room 315

Office Hours

Office hours are held by appointment, in person or remotely, during business hours (Monday–Friday). Please contact your professor or TA to arrange a meeting.

4 Topics

4.1 Introduction to Programming

- Week 1: Introduction to the Process of Software Development (3 hours)
- Week 2: Structured Programming Basics (3 hours)
- Weeks 3–6: Selections (12 hours)
- Weeks 7–10: Loops (12 hours)
- Weeks 11–13: Functions (9 hours)
- Weeks 14–16: Structs, Single- and Multidimensional Arrays (9 hours)

4.2 Object-oriented Programming

- Weeks 1–3: Objects and Classes (9 hours)
- Weeks 4–5: Exception Handling (6 hours)
- Weeks 6–8: Encapsulation, Inheritance, and Polymorphism (9 hours)
- Weeks 9–10: Abstract Classes and Interfaces (6 hours)
- Weeks 11–12: GUI and Computer Graphics Basics (6 hours)
- Weeks 13–14: Templates (6 hours)
- Weeks 15–16: Working with I/O (6 hours)

5 Learning Outcomes

By the end of this course, students will:

1. **Acquire basic programming skills to write and test programs:**
 - Write programs using fundamental C++ language constructs
 - Apply debugging and testing techniques to ensure correctness
2. **Develop proficiency in modeling object-oriented systems:**
 - Represent software designs using object-oriented principles
 - Use appropriate modeling techniques to describe classes, objects, and relationships

3. **Evaluate and use appropriate tools and techniques for software development:**
 - Use code editors, compilers, build systems, debuggers, IDEs, and other development tools effectively
 - Employ best practices in version control and collaborative development
4. **Design, plan, and critically evaluate software solutions to problems:**
 - Work with user requirements and constraints
 - Assess possible solutions for correctness and maintainability
5. **Assess systems in terms of quality attributes and trade-offs:**
 - Use automated testing and code analysis tools to evaluate software quality
 - Weigh design decisions against project constraints and overall goals
6. **Reflect on and reason about information-handling problems:**
 - Critically examine data structures and algorithms for effective information management
 - Consider the privacy, data protection, and ethical implications of software solutions
7. **Understand and apply ethical principles and professional standards:**
 - Demonstrate awareness of legal and ethical issues in computing
 - Act professionally and responsibly when designing and implementing software

6 Assignments

6.1 Moodle/e-Course Checkpoints

Students must maintain instructor-provided private GitHub repositories for their assignments. They must periodically commit and push the required number of lab solutions, as directed by the course staff. Instructors or teaching assistants will review the work either during lab sessions (on-site) or after the submission deadline (off-site) and award points based on completed assignments.

6.2 OJ Problems and Projects

Students develop a course project that applies the material to a real-world problem. Depending on the offering, they may also be assigned problems from an online judge (OJ) platform such as Codeforces to sharpen their problem-solving skills and prepare for programming contests or job interviews. Students present and defend their work to the instructor during the midterm or final examination period.

7 Course Materials, Recordings, and Screencasts

All course materials are available on GitHub at <https://github.com/auca/com.122-119>. Using the course repository will help students become familiar with the Git version control system and GitHub.

We aim to record each class and upload the recording to YouTube as time permits, but recordings are not guaranteed. If you need a recording right away, consider recording the session on your own computer. Links to YouTube recordings can be found in the course repository at <https://github.com/auca/com.122-119>.

While recordings provide flexibility, they are not a substitute for attending classes. Active participation is crucial for success in this course. Accumulating three or more unexcused absences may lead to an *X* grade. If overall class attendance is poor, the instructor reserves the right to discontinue class recordings.

Access lecture screencasts remotely via Zoom at either <http://com-122-zoom.auca.space> or <http://com-119-zoom.auca.space>. When joining a Zoom session, identify yourself with your properly capitalized first and last names in the Latin alphabet. Your lab instructor may also use Zoom or other tools for remote work and set additional etiquette rules. Ask your instructor for details. Install and configure the required tools on your computer during the first week of the semester so that you can use your camera and microphone and share your screen.

To attend lectures or labs remotely, you must request permission from the instructor in advance. Please send a brief email explaining why you need to attend remotely (e.g., illness, personal reasons) and provide appropriate documentation to support your request, either immediately or as a follow-up in the same email thread. If documentation is not provided within a reasonable time, your absence will be recorded as unexcused, and you may receive a grade of *X* if you accumulate three or more unexcused absences. Because we receive many such requests, do not expect a reply to your email. You must still submit all required documentation as soon as possible in the same thread. AUAF students who are outside the country and can only participate online do not need to request this permission.

8 Software

Students are advised to install the following software on their computers at the start of the course. Additional installations may be required later.

- Git: <https://git-scm.com/downloads>
- Visual Studio Code: <https://code.visualstudio.com>

We select tools and design tasks so that you can complete the course on any modern operating system (Windows, macOS, or Linux). However, we officially support only the operating systems used on the lab computers (Windows 10 or 11) and on the automated graders (Ubuntu 24.04). If you cannot complete an exercise on your own computer, ask the teaching staff for help with troubleshooting. If the problem cannot be resolved, use the lab machines.

9 Hardware

There are no specific hardware requirements: you can use the lab machines, where we guarantee that all course tasks can be completed. If you use your own computer, we recommend ample memory, fast disk I/O, and a GPU that supports current graphics APIs.

10 Reading

1. Introduction to Programming with C++, Third Edition (International Edition), by Y. Daniel Liang (AUCA Library Call Number: QA76.73.C153 L53 2014, ISBN: 978-0273793243)

10.1 Supplemental Reading

1. The C++ Programming Language, Fourth Edition, by Bjarne Stroustrup (AUCA Library Call Number: QA76.73.C153 S77 2013, ISBN: 978-0321563842)
2. A Tour of C++, Third Edition, by Bjarne Stroustrup (AUCA Library Call Number: QA76.73.C153 2023, ISBN: 978-0136816485)

11 Grading

The preliminary distribution of points is outlined below. This distribution may change during the course if tasks are canceled, merged, or made optional (for bonus points). Such changes usually result from software issues, online service outages, or class cancellations due to events outside our control.

Some learning management systems (LMSs), such as Moodle, may not calculate final scores and grades correctly until all tasks have been published in the system and properly weighted. Do NOT assume that your grade is accurate until all tasks are in the system and your instructor has asked you to review your scores and grades.

Attendance points do not count toward your final grade. They are used only to help us determine whether to assign *X* grades in the middle of the semester, and they are removed from the totals in Moodle at the end of the semester.

11.1 Moodle/e-Course Checkpoints

Your instructor will periodically announce reviews of your work. The maximum number of points available for each category of work is listed below.

- Lab Submissions and Classwork (16%)
- Online Judge Problems (16%, COM 122 only)
- Course Project (13% for COM 122, 29% for COM 119)

For COM 119, the Course Project consists of two types of work. Parts similar to lab assignments are completed with the instructor during the second half of the

semester and are worth 16% in total. An independent component is worth 13%. Together they make up the 29% available for the Course Project. Each lab in the first half of the semester is worth more because its tasks are more complex. As a result, lab submissions unrelated to the Course Project are still worth 16% in total.

11.2 Exams

You will take the midterm examination in the middle of the course and the final examination at the end. Both carry substantial weight in your grade.

- Midterm Exam (25%)
- Final Exam (30%)

11.3 Bonus Opportunities

You may recover points lost on other tasks by attending Extra Lecture Sessions or visiting the Writing and Academic Resource Center (WARC). Each visit earns one point, up to the limit listed below.

- Extra Lecture Sessions or WARC Bonus (+5%)

11.4 Totals

The course total of 100% consists of the Moodle/e-Course Checkpoints (45%) and the two exams (55%). You can earn up to 5% as a bonus by attending WARC sessions or Extra Lecture Sessions conducted by your instructors.

11.5 Scale

- [94%–100] %: A
- [90%–94) %: A-
- [87%–90) %: B+
- [83%–87) %: B
- [80%–83) %: B-
- [77%–80) %: C+
- [73%–77) %: C
- [70%–73) %: C-
- [67%–70) %: D+
- [63%–67) %: D
- [60%–63) %: D-
- Less than 60%: F

Requests for a higher grade because a score falls marginally short of a threshold will be ignored. A score of 93.99 is an A-, NOT an A. Requests for extra assignments to earn additional points will also be disregarded.

12 Rules

In addition to the rules in this syllabus, students are required to follow the Code of Conduct of the American University of Central Asia.

12.1 Participation

Active participation in class may be rewarded with extra points at the instructor's discretion. Poor performance in class may result in points being deducted from the final grade.

Instructors may conduct unannounced checks during class, so students **MUST** be prepared for every session to avoid losing points. Students who miss a class with graded work without a valid reason will also lose points, unless the absence is due to force majeure. To avoid losing points, inform the instructors of the reason for an absence in advance.

You can attend WARC consultation sessions run by student tutors or Extra Lecture Sessions led by the instructors. Each session you attend earns one extra point, up to five points in total. At the end of the course, we collect the attendance reports and add these points to your course total when calculating your final grade.

12.2 Questions

A question from one student is often relevant to others as well. Students are therefore encouraged to post questions publicly on the discussion board of the course's LMS, such as the AUCA e-Course System, so that all students can benefit from the discussion.

Students must not post the complete source code for any task on the LMS discussion board. Any such public post results in a grade of zero for that assignment. Students should also avoid vague or overly general questions such as "Why doesn't my code work?" and are expected to analyze and debug their code carefully before asking for help.

12.3 Late Policy

Late submissions and late exams are not accepted. Exceptions may be granted at the professor's discretion only in cases of force majeure. If you fall ill, face serious personal difficulties, or have technical problems with your computer or internet connection, you must notify the instructors at least 24 hours before the deadline. Otherwise, no extension will be granted. Requests made on the day of the deadline will be treated as procrastination and will not be considered. No student is granted more than one emergency extension during the course.

During the six hours before any assignment deadline, the instructors enter silent mode: they do not answer assignment-related questions or consider requests for office hours. At all other times, the instructors make every effort to answer questions and provide help, whether via Zoom or in person during office hours.

12.4 Exam and Task Submission Ceremonies

Students must follow all exam and task submission protocols, which means strictly adhering to every rule the instructors specify, whether in writing or verbally. Failure to do so results in a loss of points.

Throughout your career, you will work with supporting documents such as contracts and timesheets. Beyond programming, you will also use project management systems, application lifecycle management tools, version control, and continuous integration and deployment pipelines. You will manage complex configurations, collaborate with colleagues by following specific methodologies, and share code through pull requests or other means. In all these processes, proper etiquette and attention to detail are essential.

Our courses aim to help you build these skills. We deduct points for not following submission rules or protocols, and we may refuse to accept an exam defense or a submitted task altogether. We also award zero points for missed task deadlines and for violations of strict exam timing rules, such as arriving late or failing to submit exam documents on time.

12.5 Administrative Drop

Instructors reserve the right to drop a student from the course for nonattendance. If a student misses three or more classes without an acceptable excuse, the faculty may assign an *X* grade and remove the student from the course.

An absence can be excused only if you complete both of the steps below and we accept your explanation.

First, write to the instructor of the class you will miss, your lecture instructor for a lecture or your lab instructor for a lab. Send a short message giving the reason, and send it **BEFORE** the class. This deadline is strict: a message that arrives after the class has taken place will not be processed, and the absence will be counted as unexcused.

Second, once your health or your situation allows, you **MUST** upload documents supporting your explanation to the same email thread. These can be medical records, a letter from your parents, or documents from an organization or an official body, as long as they confirm that the situation was real and that it prevented you from attending. Upload them as soon as you can, since we review the threads before *X* grades are decided and cannot take later material into account.

If the situation was genuinely serious and your documents are valid, the absence may be excused. We decide each case on its own merits, and we may ask you or your guardians to contact us or to meet with us at AUCA so that the documents can be verified. We may also decline to excuse an absence if the circumstances do not warrant it, or if the documents raise doubts.

Do not expect a reply to these threads. We receive too many messages to answer them or even confirm that they arrived, so you will usually hear from us only if we need something more from you. Every thread is still collected and read carefully, so silence does not mean your message was lost or ignored. For the same reason, we do not accept appeals or negotiations about these decisions: handling them would take time away from our core work.

12.6 Incomplete Grade

Consistent with the late exam policy, a grade of *I* (Incomplete) is granted only under highly exceptional circumstances. Students must request an *I* grade well in advance of the final exam. Requests made the day before the final exam or later will not be considered.

12.7 Academic Honesty

Plagiarism is the act of copying or appropriating someone else's words or ideas and presenting them as one's own. In this course, plagiarism can involve any kind of work, including program code, comments, software documentation, design specifications, requirements documents, project reports, and technical analyses.

Examples of plagiarism in a software engineering course include:

- Submitting code written by others or generated by AI as your own
- Modifying existing code or designs (e.g., changing variable names or restructuring code) and claiming originality
- Incorporating code snippets, sentences, design patterns, or intellectual content from any source without citation
- Using algorithms, patterns, or architectural designs without proper acknowledgment
- Using graphics, datasets, audio, video, or other materials from external works without attribution
- Purchasing or otherwise acquiring code, software, or project content and presenting it as original work

Plagiarism is unethical and undermines the educational process. The consequences are as follows:

- First instance: A grade of zero for the plagiarized work and a report filed with the Registrar's Office
- Second instance: A failing grade (*F*) for the entire course

The student who plagiarizes and the student whose work was copied face the same consequences because every student is responsible for safeguarding their assignments, code, and related materials against unauthorized use. This policy reflects professional expectations in software engineering, where failing to protect intellectual property can lead to dismissal, reputational damage, compromised customer and user safety, and even lawsuits.

All work in this course must be completed individually. Group work, collaboration with peers, and seeking help from other students are strictly prohibited. Relying on classmates often spreads misunderstandings and increases the risk of academic dishonesty. Students may seek help only from instructors, teaching assistants (TAs), and WARC tutors, so that guidance comes from qualified sources and each student develops their own skills and understanding.

The use of artificial intelligence, including generative AI tools, to complete any assignment, project, or exam, whether off-site or on-site, is strictly prohibited. A student suspected of using AI assistance will be required to perform a similar task and answer related questions in a supervised setting with the course instructors. Failure to complete this task satisfactorily or to answer the questions convincingly will lead to the conclusion that the work was AI-generated, and the Academic Honesty policies above will be enforced, including possible disciplinary action.

This course also follows all university-wide AUCA policies on academic honesty. Where university regulations are stricter than the rules stated here, the stricter rules apply.

12.8 Use of AI by the Course Staff

The instructors and teaching assistants for this course use artificial intelligence tools in their work. The instructors design the course, develop its content, plan instruction and assessment, and make all academic decisions themselves. They use AI for routine tasks: organizing and formatting materials, proofreading, keeping course documents consistent, and finding possible errors or inconsistencies in grading for their own review.

A person checks every grade before it is recorded. Practice work, including labs and projects, may be graded automatically or prechecked by AI, but a teaching assistant or the instructor verifies every automated grading result before it counts toward the course grade. Midterm and final examinations are graded by the instructor alone. AI may be used to precheck exam submissions, but neither AI nor a teaching assistant assigns an exam grade: the instructor reviews each submission and makes each grading decision personally, one by one, before the result is recorded.

The rules for student use of AI are set out in the Academic Honesty section.

13 Access and Support Services

We are committed to providing an inclusive learning environment that supports the needs of all students. If you have a disability or need specific accommodations to participate fully, contact the instructors as early as possible. We will work with you to make appropriate adjustments.

For guidance on time management, presentation skills, writing, or study strategies beyond the scope of software engineering, contact the AUCA Advising Office, which offers practical workshops and peer advising to help you with academic challenges.

If you would like help from a student tutor, you can schedule an appointment through the AUCA Writing and Academic Resource Center (WARC). Tutoring is available in person and online, and additional resources are listed on the WARC webpage. Note, however, that team or group work with other students is not permitted on assignments in this course.

If you feel stressed or overwhelmed or are struggling with emotions, relationships, or other mental health concerns, we encourage you to seek support from the AUCA Counseling Service, which provides free, confidential professional help to students.